

Practical WS20: Behaviour of a Cantilever

Time: 1 hr 15 min

Name: **Suggested Solution** ()

Class: **26S0**

Date:

[2023/P4/Q3 Modified]

A cantilever beam is a beam that is supported at one end only.

In this experiment, you will investigate how the following properties affect the behaviour of a cantilever beam:

- the length of the beam
- the load on the beam
- the cross-section of the beam
- the stiffness of material from which the beam is made.

(a) Set up the half-metre rule on a flat surface such as benchtop or a stool, as shown in Fig. 1.1.

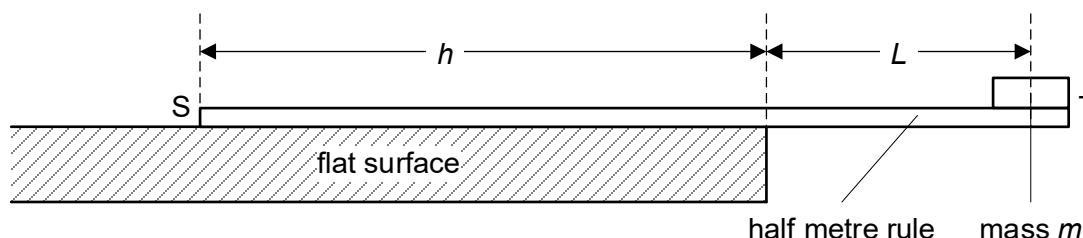


Fig. 1.1

The ends of the rule are S and T.

Use Blu-Tack to firmly attach a mass m , where $m = 50 \text{ g}$, at T.

The distance between S and the edge of the surface is h .

The distance between the centre of mass m and the edge of the surface is L .

(i) Adjust the position of the rule so that S just lifts off the surface.

Measure and record h and L .

$$h = (35.5 + 35.5)/2 = 35.5 \text{ cm}$$

$$L = (13.0 + 13.0)/2 = 13.0 \text{ cm}$$

Record h and L to the nearest mm.

Repeated readings and taking average.

$48 \text{ cm} \leq h + L \leq 49 \text{ cm}$. No mark if $h + L = 50 \text{ cm}$.

$$h = 35.5 \text{ cm}$$

$$L = 13.0 \text{ cm} \quad [1]$$

- (ii) Calculate $\frac{h}{L}$.

$$\frac{h}{L} = \frac{35.5}{13.0} = 2.73$$

Correct calculation and s.f. (3 s.f.). Ratio varies due to different rulers.

$$\frac{h}{L} = 2.73$$

[1]

- (iii) Estimate the percentage uncertainty in your value of $\frac{h}{L}$.

$$\text{Percentage uncertainty of } \frac{h}{L} = \left(\frac{\Delta h}{h} + \frac{\Delta L}{L} \right) \times 100\% = \left(\frac{0.3}{35.5} + \frac{0.4}{13.0} \right) \times 100\% = 3.9\%$$

$0.3 \text{ cm} \leq \Delta h < \Delta L \leq 0.5 \text{ cm}$. [1]

Correct calculation and s.f. (1 or 2 s.f.) [1]

Note: $\Delta L > \Delta h$ due to difficulty in locating the centre of mass of the slotted mass accurately.

Candidates were asked to estimate the percentage uncertainty in their value of h/L . Many candidates successfully added $\Delta h/h$ and $\Delta L/L$ with very few dividing or subtracting. They also realised that the range of uncertainties for the Δh and ΔL were not the same. After multiplying the sum by 100, some candidates went on to either multiply or divide by the value of h/L . Some candidates mixed up mm and cm for the uncertainties, and some divided the sum of the uncertainties by the corresponding value of h/L .

$$\text{percentage uncertainty of } \frac{h}{L} = 3.9\%$$

[2]

- (b) Vary m and repeat (a)(i) and (a)(ii).

m / g	h_1 / cm	h_2 / cm	$\langle h \rangle / \text{cm}$	L_1 / cm	L_2 / cm	$\langle L \rangle / \text{cm}$	$\frac{h}{L}$
50	35.5	35.5	35.5	12.8	13.1	13.0	2.73
100	39.4	39.5	39.5	9.0	8.9	9.0	4.4
150	41.5	41.9	41.7	6.8	6.8	6.8	6.1
200	43.1	43.3	43.2	5.4	5.2	5.3	8.2
250	43.9	43.8	43.9	4.5	4.6	4.6	9.5

[2]

Correct headings for m , h , L and h/L with appropriate units.

Correct d.p. for h and L . [1]

Correct calculation and s.f. for h/L column. [1]

- (c) It is suggested that h , L and m are related by the expression:

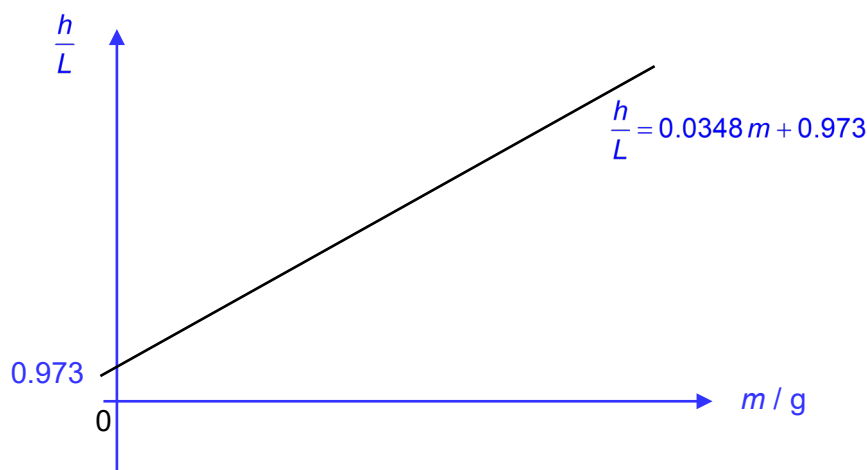
$$\frac{h}{L} = \frac{2m}{X} + 1$$

where X is the mass of the half-metre rule.

- (i) Plot a suitable graph using a spreadsheet.

Sketch the graph in the space below.

Plot the graph of h/L vs m , where the gradient is $2/X$.



[1]

- (ii) Use your graph to determine a value for X .

$$\frac{2}{X} = 0.0348$$

$$X = 57.5 \text{ g}$$

$$X = \underline{57.5 \text{ g}} \quad [1]$$

- (d) Use a balance to determine the mass X of your half-metre rule.

$$X = \underline{68.03 \text{ g}}$$

Show whether your value in (a)(iii) explains the difference between your two values of X .

$$\text{Percentage difference in } X = \frac{68.03 - 57.5}{68.03} \times 100\% = \frac{10.53}{68.03} \times 100\% = 15\%$$

Since the percentage difference in X (15%) is greater than the percentage uncertainty in h/L (3.9%), (a)(iii) does not explain the difference between the two values of X .

[1]

X in (d) was measured directly using a more accurate instrument.

Hence, the difference must reference this value as the base.

Correct conclusion upon comparison with value in (a)(iii).

(e) CAUTION! Handle the blade with care.

You will now use a metal hacksaw blade as a cantilever beam.

When one end of the blade is fixed and the other end is loaded, the loaded end will move downwards, as shown in Fig. 1.2.

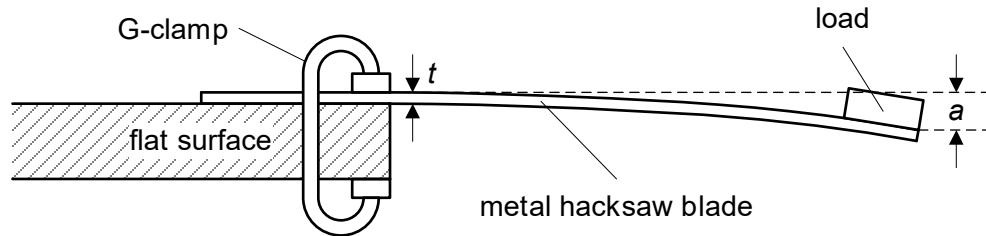


Fig. 1.2

The distance moved down is a , given by:

$$a = \frac{4MgL^3}{Yut^3}$$

where:

- M is the mass of the load
- L is the distance between the centre of the load and the edge of the surface
- u is the width of the blade
- t is the thickness of the blade
- Y is the stiffness of the metal
- $g = 9.81 \text{ N kg}^{-1}$

(i) Wearing safety goggles, set up the apparatus as shown in Fig. 1.2.

Use Blu-Tack to firmly attach a 50 g mass to the metal hacksaw blade.

Take measurements to determine a value for Y .

$$t = \frac{1}{2}(0.732 + 0.721) = 0.727 \text{ mm} = 0.000727 \text{ m}$$

$$u = \frac{1}{2}(12.396 + 12.284) = 12.340 \text{ mm} = 0.012340 \text{ m (using digital micrometer screw gauge)}$$

$$\text{or } u = \frac{1}{2}(0.012 + 0.012) = 0.012 \text{ m (using metre rule to measure)}$$

$$L = 0.268 \text{ m}$$

$$a = \frac{1}{2}(0.080 + 0.079) = 0.080 \text{ m}$$

$$Y = \frac{4MgL^3}{aut^3} = \frac{4 \times 0.050 \times 9.81 \times 0.268^3}{0.080 \times 0.012340 \times 0.000727^3} = 9.96 \times 10^{10} \text{ Pa}$$

Repeated readings for a , t and u . [1]

Correct d.p. and units for t , u (either ruler or MSG), L and a . [1]

$0.7 \text{ mm} \leq t \leq 0.8 \text{ mm}$, $1.1 \text{ cm} \leq u \leq 1.3 \text{ cm}$ [1]

Correct calculation of Y , conversion from Pa to GPa and s.f. (2 or 3 s.f.) [1]

$$Y = 99.6 \text{ GPa or } 100 \text{ GPa} \quad [4]$$

(ii) Suggest **one** significant source of uncertainty in this experiment.

1. The metre rule may not be vertical as it is handheld thus affecting the accuracy of a .
2. The end of the hacksaw blade is deflected even without mass attached. If this is not accounted for, a will not be accurate due to systematic error.

[1]

Any reasonable significant source of error. However, just stating "parallax error" without linking to the context of the experiment is not accepted.

Reject effect of mass of Blu-tack as it is insignificant compared to the mass of the load and the blade.

(iii) The value of Y for wood is 12 GPa.

Explain why it is **not** possible to perform this experiment using a wooden beam with the same dimensions as the metal hacksaw blade.

The stiffness Y of wood (12 GPa) is 8.3 times less than that of the hacksaw blade (100 GPa).

According to $a = \frac{4MgL^3}{Yut^3}$, the end deflection a of the wooden beam is 8.3 times that of the metal which may not be possible without breaking.

[1]

For wooden beam of the same dimensions,

$$a = \frac{4MgL^3}{Yut^3} = \frac{4 \times 0.050 \times 9.81 \times 0.268^3}{12.0 \times 10^9 \times 0.012340 \times 0.000727^3} = 0.664 \text{ m}$$

The calculated deflection (0.664 m) of the wooden beam would exceed the length of beam (0.314 m). Hence, the wooden beam will most likely break when loaded with 50 g. Accept reasonable presentation or argument.

- (f) A diving board at a swimming pool has length L . A person of mass M jumps up and down on the end of the diving board.

The frequency of oscillation of the end of the diving board is f , given by:

$$f^2 = \frac{Z}{ML^3}$$

where Z is a constant.

The metal hacksaw blade can be used as a model diving board.

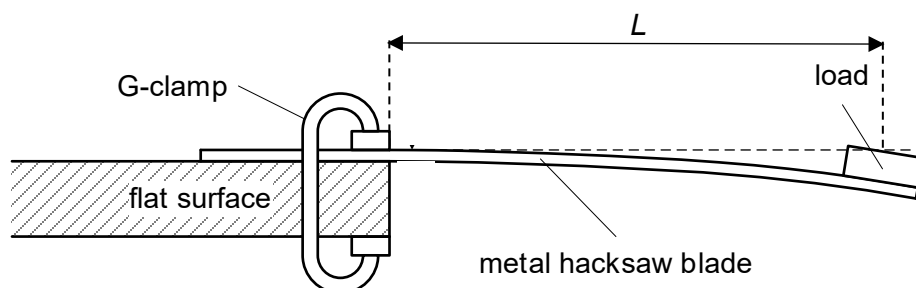
Plan an investigation to find Z for the blade.

Your answer should include a diagram, your experimental procedure and a precaution to improve the safety of the experiment.

Wearing safety goggles, use your apparatus to determine **two** values of Z for different values of M and L . Tabulate your results.

CAUTION! Candidates should use no more than two 50 g masses.

L / m	M / g	N	Time for N oscillations			T / s	f / Hz	$Z / \text{kg m}^3 \text{ s}^{-2}$
			t_1 / s	t_2 / s	$\langle t \rangle / \text{s}$			
0.268	50	40	22.54	22.50	22.52	0.5630	1.776	3.0×10^{-3}
0.258	100	40	26.94	27.01	26.98	0.6745	1.483	3.77×10^{-3}



- Set up the apparatus as shown in the figure above.
- Attach the load M to the end of the blade.
- Measure and record the length L with a metre rule.
- Push down the blade by a small displacement and release.
- Measure and record the time t taken for N oscillations with a stopwatch.
- The period of oscillations of the blade is $T = t/N$ and frequency is $f = N/t$ or $f = 1/T$.
- Repeat steps 2 to 6 by varying M and L for 8 sets of readings by attaching more mass and adjust length of blade (or adjusting one and keeping the other constant).

8. Plot a graph of f^2 against $1/ML^3$. Determine Z from the gradient of the graph.

9. Wear covered shoes to prevent foot injury if masses fall. Or, wear gloves when handling hacksaw blade to prevent cuts to the hand.

[7]

Table [2]:

2 sets of different L and M tabulated. [1]

Correct calculation of Z with appropriate s.f. and units. [1]

Planning [5]:

Diagram with L labelled. [1]

Method to measure length L and time t . [1]

Method of varying M and L for at least 8 total readings (allow varying one variable while keeping the other constant). [1]

State quantities to plot and method to determine Z . [1]

One reasonable safety. Wearing safety goggles is not accepted as it was already mentioned in the question [1]

[Total: 22]